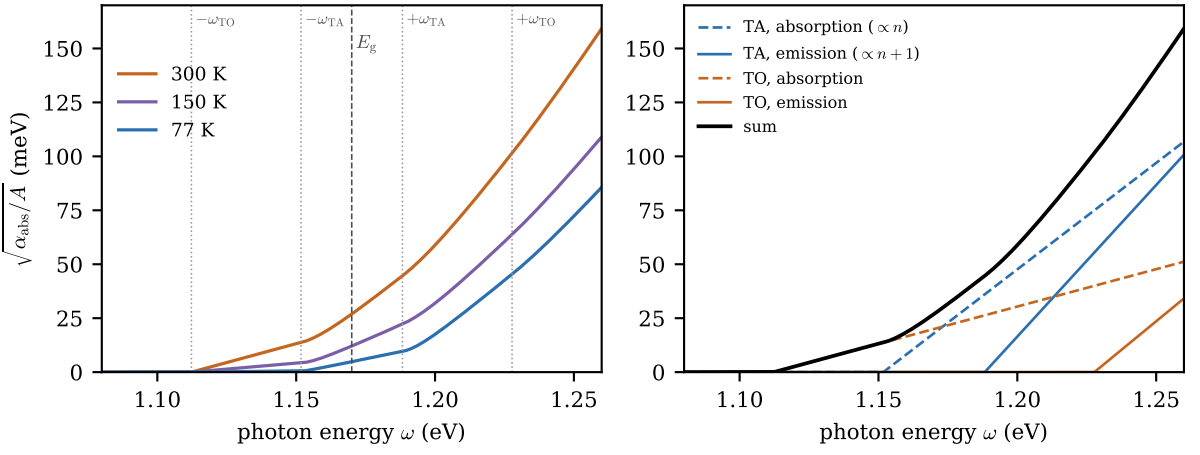
(a) sum of all components
$-\omega_{\mathrm{TO}}$
$-\omega_{\mathrm{TA}}$
E_{g}
$+\omega_{\mathrm{TA}}$
$+\omega_{\mathrm{TO}}$
300 K
150 K
77 K
$\sqrt{\alpha_{\mathrm{abs}}/A}$ (meV)
0
25
50
75
100
125
150
1.10
1.15
1.20
1.25
photon energy ω (eV)
(b) components at 300 K
TA, absorption ($\propto n$)
TA, emission ($\propto n+1$)
TO, absorption
TO, emission
sum
0
25
50
75
100
125
150
1.10
1.15
1.20
1.25
photon energy ω (eV)